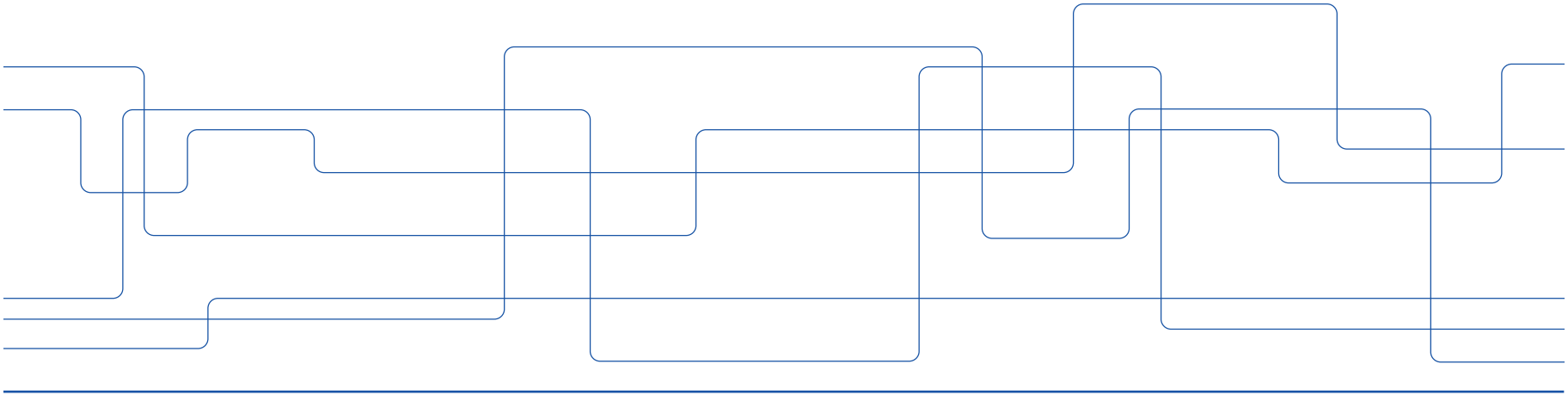
# MH2000 Experimental Methods

## Thermodynamic Measurements

Dr. Sudhanshu Kuthe



# Thermodynamic measurements

- Lecture 1: Isopiestic/Isobaric methods, transpiration method, gas mixtures for controlled chemical potentials, liquid–liquid chemical equilibria

Room: B25

- Lecture 2: Gibbs energy function (gef), second- and third-law methods, electrochemical techniques

Room: D33

# Thermodynamic measurements

- ❑ Lecture 1: Isopiestic/Isobaric methods, transpiration method, gas mixtures for controlled chemical potentials, liquid–liquid chemical equilibria

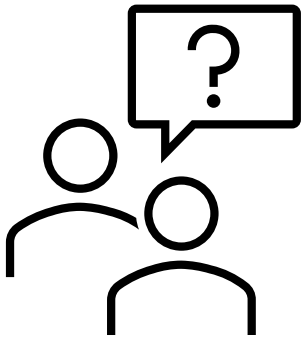
Room: B25

- ❑ Lecture 2: Gibbs energy function (gef), second- and third-law methods, electrochemical techniques

Room: D33

**Please Scan**

**“How can we determine Gibbs energy, enthalpy, entropy and chemical activity from experimental measurements?**



# From Experimental Measurements to Gibbs Energy

**Lecture 1: Establish equilibrium**

Measure vapor pressure, gas composition and phase composition

**Equilibrium constants**

$$\Delta G^\circ = -RT \ln K$$

**Cell voltage**

$$\Delta_r G = -zFE$$

**Determine thermodynamic properties**

Gibbs energy · Enthalpy · Entropy · Chemical activity

**If we know the Gibbs energy of a reaction at room temperature, can we predict it accurately at steelmaking temperatures (1600°C)?**

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

The goal of every method today: find $\Delta G^\circ$ (and $\Delta H^\circ$, $\Delta S^\circ$) of a reaction

## A Look it up

**Gibbs energy function (gef)**

Precise $\Delta G^\circ$ at any T from tabulated data

$$G = H - TS$$

$$H(T) = H(298) + \int_{298}^{T} C_p \, dT$$

$$S(T) = S(298) + \int_{298}^{T} \frac{C_p}{T} \, dT$$

Therefore, accurate Gibbs energy calculations require heat-capacity information.

The Gibbs energy function provides a convenient way to organize this information in thermodynamic tables.

# Gibbs energy function (gef)

- For very precise thermodynamics calculations, following formulas for Gibbs energy are not precise enough:

$$\Delta G°_T = A + BT\ln T + CT^2 + DT^{-1} + ET$$

$$\Delta G°_T = A + BT\ln T + CT$$

$$\Delta G°_T = A + BT$$

  - Curve fitting of $c_p = a + bT + cT^{-2} + dT^2$ leads to errors

  - Would be most precise to tabulate $c_P$ values of the reactants and products and then integrate

# Gibbs energy function (gef)

- There is also the 298 K Gibbs energy function, $\text{gef}_{298}(T)$
  - When low-temperature heat capacity data is missing

$$\frac{\Delta G^\circ_T}{T} = \frac{\Delta\left(G^\circ_T - H^\circ_{298}\right)}{T} + \frac{\Delta H^\circ_{298}}{T}$$

- Example, calculate $\Delta G^\circ_{1000}$ for the following reaction

$$4\text{Cu(s)} + \text{O}_2\text{(g)} = 2\text{Cu}_2\text{O(s)}$$

# Gibbs energy function (gef)

- Example (continued)

$$4Cu(s) + O_2(g) = 2Cu_2O(s)$$

  - Use thermodynamic tables, NIST/JANAF for example

Enthalpy Reference Temperature = $T_r$ = 298.15 K  Standard State Pressure = $p^\circ$ = 0.1 MPa

**Copper Oxide ($Cu_2O$)** **$Cu_2O_1$(cr)**

| $T$/K | $C_p^\circ$ | $S^\circ$ | $-[G^\circ-H^\circ(T_r)]/T$ | $H^\circ-H^\circ(T_r)$ | $\Delta_f H^\circ$ | $\Delta_f G^\circ$ | log $K_f$ |
|---|---|---|---|---|---|---|---|
| | J·K⁻¹mol⁻¹ | | | kJ·mol⁻¹ | | | |
| 0 | 0. | 0. | INFINITE | −12.600 | −168.952 | −168.952 | INFINITE |
| 100 | 39.499 | 37.092 | 141.510 | −10.442 | −169.584 | −162.621 | 84.944 |
| 200 | 53.619 | 69.143 | 97.826 | −5.737 | −170.360 | −155.348 | 40.573 |
| 298.15 | 62.543 | 92.360 | 92.360 | 0. | −170.707 | −147.886 | 25.909 |
| 300 | 62.664 | 92.747 | 92.361 | 0.116 | −170.709 | −147.745 | 25.725 |
| 400 | 67.688 | 111.531 | 94.888 | 6.657 | −170.641 | −140.092 | 18.294 |
| 500 | 70.922 | 127.000 | 99.809 | 13.596 | −170.363 | −132.484 | 13.840 |
| 600 | 73.448 | 140.161 | 105.465 | 20.818 | −169.961 | −124.944 | 10.877 |
| 700 | 75.645 | 151.651 | 111.259 | 28.274 | −169.479 | −117.478 | 8.766 |
| 800 | 77.666 | 161.886 | 116.959 | 35.941 | −168.929 | −110.087 | 7.188 |
| 900 | 79.493 | 171.141 | 122.473 | 43.801 | −168.325 | −102.767 | 5.964 |
| 1000 | 81.245 | 179.606 | 127.769 | 51.837 | −167.688 | −95.517 | 4.989 |
| 1100 | 83.385 | 187.443 | 132.842 | 60.061 | −167.029 | −88.331 | 4.194 |
| 1200 | 86.110 | 194.812 | 137.702 | 68.532 | −166.322 | −81.208 | 3.535 |
| 1300 | 89.173 | 201.824 | 142.367 | 77.294 | −165.608 | −74.144 | 2.979 |
| 1400 | 92.489 | 208.552 | 146.855 | 86.375 | −191.165 | −66.321 | 2.474 |
| 1500 | 95.987 | 215.052 | 151.186 | 95.798 | −190.132 | −57.438 | 2.000 |
| 1516.700 | 96.587 | 216.118 | 151.895 | 97.406 | ---- CRYSTAL <--> LIQUID ---- | | |
| 1600 | 99.631 | 221.362 | 155.376 | 105.578 | −188.755 | −48.635 | 1.588 |
| 1700 | 103.380 | 227.514 | 159.439 | 115.728 | −187.020 | −39.928 | 1.227 |
| 1800 | 107.212 | 233.532 | 163.389 | 126.257 | −184.917 | −31.335 | 0.909 |
| 1900 | 111.115 | 239.433 | 167.236 | 137.173 | −182.440 | −22.868 | 0.629 |
| 2000 | 115.068 | 245.232 | 170.992 | 148.481 | −179.581 | −14.542 | 0.380 |

PREVIOUS: June 1966  CURRENT: December 1977

**Copper Oxide ($Cu_2O$)** **$Cu_2O_1$(cr)**

1023

NIST-JANAF THE

# Gibbs energy function (gef)

- Example (continued) $4Cu(s) + O_2(g) = 2Cu_2O(s)$

| Species | $-\frac{G°_T - H°_{298}}{T}$ | $\Delta H°_{298}$ |
|---|---|---|
| Cu | 46.261 | 0 |
| $O_2(g)$ | 220.875 | 0 |
| $Cu_2O(s)$ | 127.769 | -167 688 |

$$-\frac{\Delta\left(G°_T - H°_{298}\right)}{T} = \left(2\times 127.769 - 220.875 - 4\times 46.261\right) = 150.381$$

$$\frac{\Delta H°_{298}}{1000} = \frac{\left(2\times -167688 - 0 - 0\right)}{1000} = -334.576$$

# Gibbs energy function (gef)

- Example (continued) $4Cu(s) + O_2(g) = 2Cu_2O(s)$

$$\frac{\Delta G^\circ_{1000}}{1000} = 150.381 - 334.576$$

$$\Delta G^\circ_{1000} = (150.381 - 334.576) \times 1000$$

$$\Delta G^\circ_{1000} = -184\ 195\ \text{J}$$ Answer

- Compare with data from Gaskell (A + B*T*):

$$\Delta G^\circ = -324400 + 138.5T = -185\ 900\ \text{J}$$

# A · Why a Gibbs energy function?

## Fitted formula

$\Delta G° = A + BT$

\+ Quick to use

− Approximate; errors grow outside the fitted range

## Integrate $c_p$ tables

$\Delta G° = \Delta H° - T\Delta S°$, with $\int c_p \, dT$

\+ Exact

− Long calculation for every reaction

The Gibbs energy function gives **the precision of the tables** with **the ease of a formula**.

# A · What the Gibbs energy function is

## Definition, for each substance

$$\mathrm{gef} = -(G^\circ_T - H^\circ_{298}) / T$$

unit: J/(mol·K)

1. **Changes slowly with T**
   so it is easy to tabulate and interpolate
2. **Listed for each substance**
   in tables such as NIST-JANAF
3. **No low-T $c_p$ data?**
   use gef relative to 298 K, as here

## Recipe, for a reaction

$\Delta\mathrm{gef} = \Sigma\ \mathrm{gef(products)} - \Sigma\ \mathrm{gef(reactants)}$

$$\Delta G^\circ_T = \Delta H^\circ_{298} - T \cdot \Delta\mathrm{gef}$$

Only two numbers per substance are needed: gef at T and $\Delta H^\circ_{298}$.

# A · Worked example: Cu oxidation at 1000 K

$$4Cu(s) + O_2(g) = 2Cu_2O(s)$$

| Species | gef at 1000 K J/(mol·K) | ΔH°f,298 kJ/mol |
|---|---|---|
| Cu(s) | 46.261 | 0 |
| $O_2$(g) | 220.875 | 0 |
| $Cu_2O$(s) | 127.769 | −167.688 |

Data: NIST-JANAF tables

**1 Combine gef**

$\Delta gef = 2(127.769) - 4(46.261) - 220.875$

$\mathbf{= -150.381\ J/K}$

**2 Enthalpy**

$\Delta H°_{298} = 2(-167.688) - 0 - 0$

$\mathbf{= -335.376\ kJ}$

**3 Put together**

$\Delta G°_{1000} = -335.376 - 1000(-0.150381)$

$\mathbf{= -185.0\ kJ}$

$\mathbf{\Delta G°_{1000} \approx -185\ kJ}$

Check: Gaskell (A + BT) gives −185.9 kJ

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K$$

The goal of every method today: find $\Delta G^\circ$ (and $\Delta H^\circ$, $\Delta S^\circ$) of a reaction

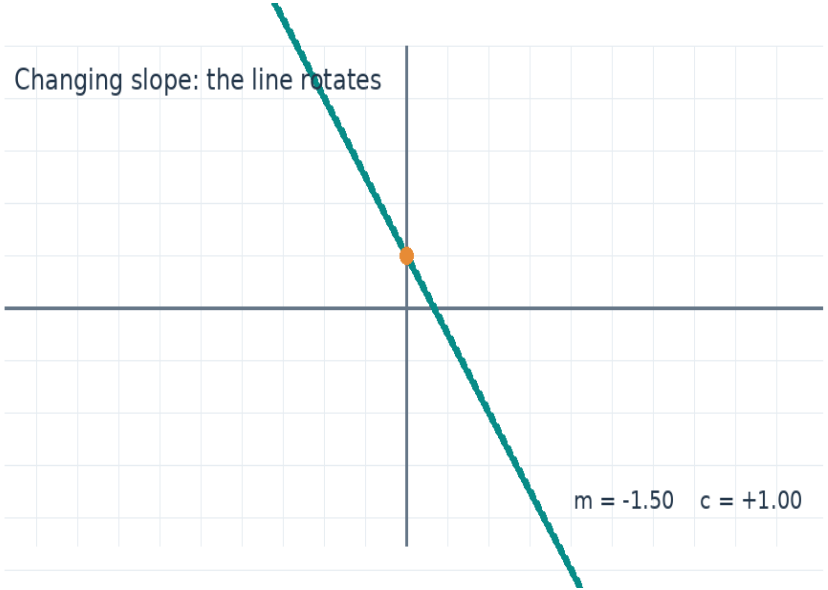
**B** **Measure K vs T**

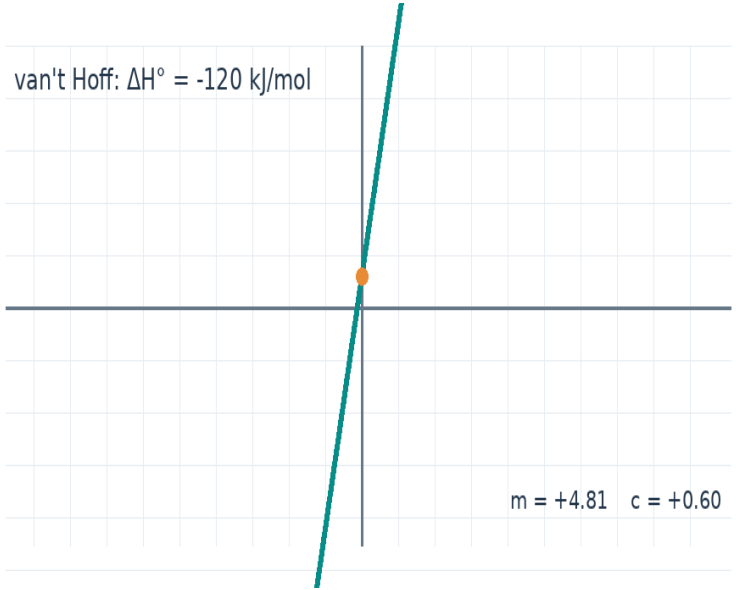
**Second- and third-law methods**

$\Delta H^\circ$ and $\Delta S^\circ$ from gas equilibria

# y = mx + c

$$\ln K = [-\Delta H^\circ/R](1/T) + \Delta S^\circ/R$$



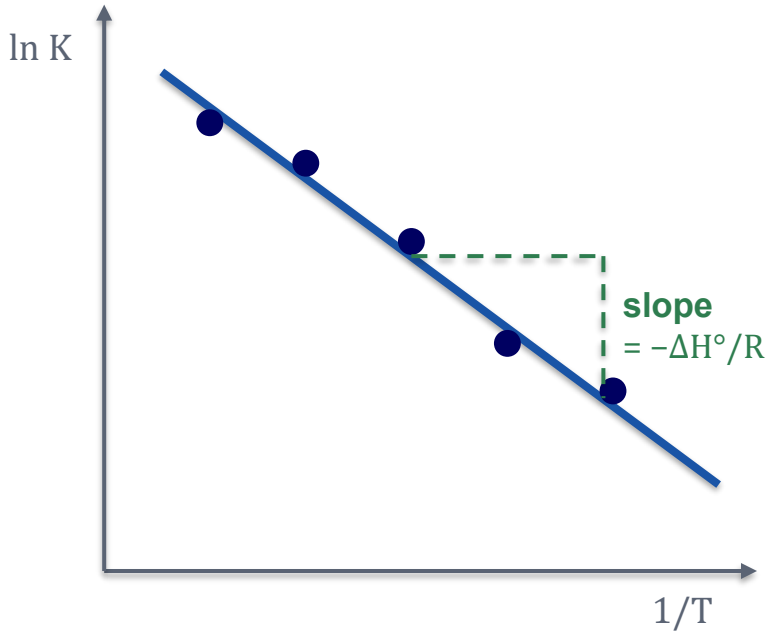

$y = \ln K \quad | \quad x = 1/T \quad | \quad m = -\Delta H^\circ/R \quad | \quad c = \Delta S^\circ/R$ (assuming constant $\Delta H^\circ$)

# B · Second-law method: K at several T

$$\ln K = -\Delta H^\circ/R * 1/T + \Delta S^\circ/R$$

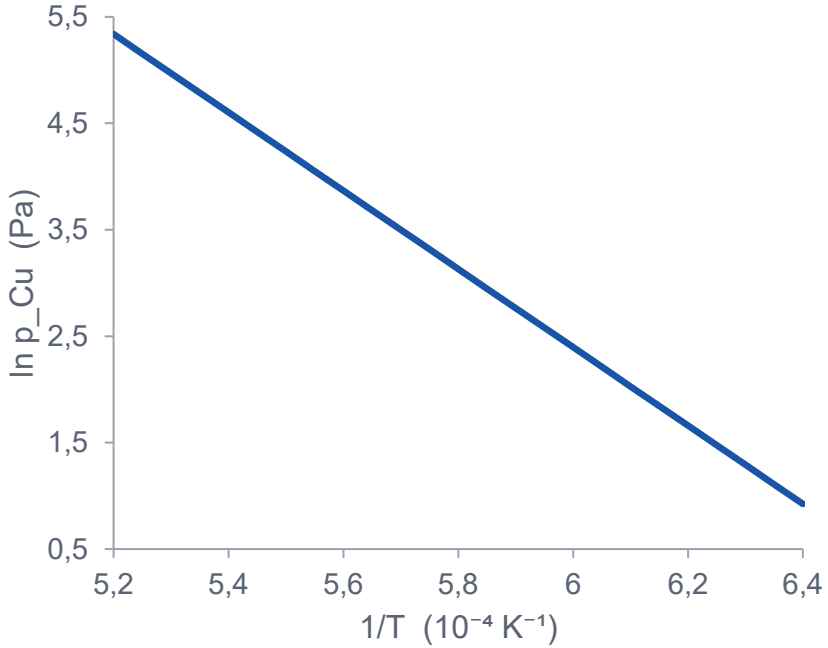
van ’t Hoff equation

| | | |
|---|---|---|
| **y** | $\ln K$ | measured |
| **x** | $1/T$ | measured |
| **slope m** | $-\Delta H^\circ/R$ | → **ΔH°** |
| **intercept c** | $\Delta S^\circ/R$ | → **ΔS°** |




Result: mean ΔH° and ΔS° over the T range

# B · Worked example: Cu vapour pressure




Fitted line from Zellars, Payne & Morris, Trans. Met. Soc. 1959

$$Cu(l) = Cu(g) \qquad K = p_{Cu}$$

$$\ln K = -3.679\times10^{4}\ /T + 24.47$$

**1** **Slope → enthalpy**

$\Delta H° = 3.679\times10^{4} \times 8.314 =$ **305.9 kJ/mol**

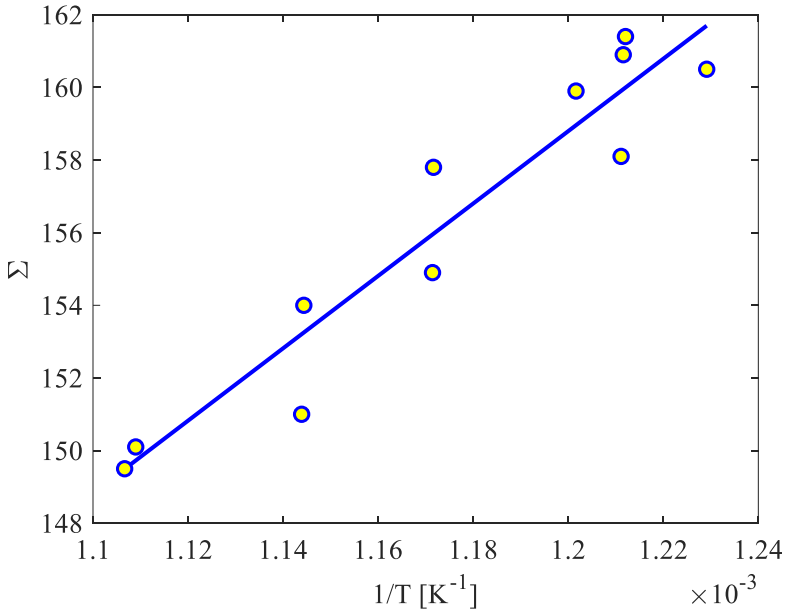
**2** **Intercept → entropy**

$\Delta S° = 24.47 \times 8.314 =$ **203.4 J/(mol·K)**

**Mean values at ≈ 1747 K**

# B · Sigma method: adding a $c_p$ correction

| | |
|---|---|
| **Problem** | ΔH° and ΔS° are not truly constant: $c_p$ changes with T |
| **Fix** | Combine ln K with the $c_p$ terms into one quantity, Σ |
| **Plot** | Σ against 1/T is again a straight line: slope $H_0$, intercept I |
| **Result** | A full expression ΔG° = A + BT ln T + CT² + D/T + ET |




Example: $HgSO_4 \cdot HgO$ dissociation

Full derivation: see appendix

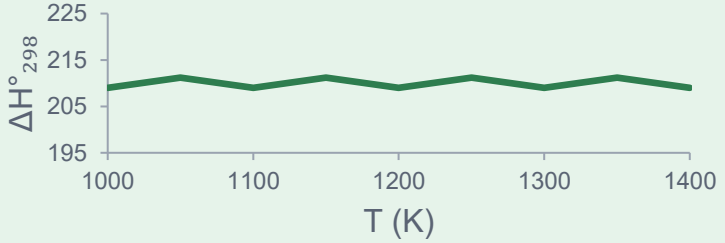
# B · Third-law method: one point is enough

$$\Delta H^\circ_{298} = -RT \ln K + T \cdot \Delta gef$$

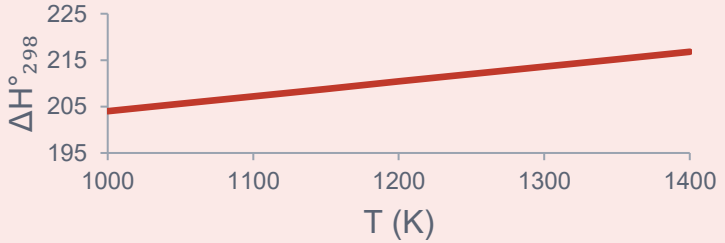
from $\Delta G^\circ T = \Delta H^\circ_{298} - T\Delta gef = -RT \ln K$

1. **Measure:** K at one temperature
2. **Look up:** Δgef at that temperature
3. **Calculate:** $\Delta H^\circ_{298}$ from every single point

**No trend** — **Data consistent ✓**




**Trend with T** — **Systematic error!**




Schematic

# Third-law method

- Third-law method (a.k.a. method of entropies) Combination of measurement of reaction equilibrium and values of gef for the reactants and products
- Since

$$\frac{\Delta G^\circ_T}{T} = \frac{\Delta\left(G^\circ_T - H^\circ_{298}\right)}{T} + \frac{\Delta H^\circ_{298}}{T} = -R\ln K$$

$$\frac{\Delta H^\circ_{298}}{T} = -R\ln K - \frac{\Delta\left(G^\circ_T - H^\circ_{298}\right)}{T}$$

- Or

$$\Delta H^\circ_{298} = -RT\ln K - T\Delta\text{gef}$$

# Third-law method

- Third-law method (a.k.a. method of entropies) Combination of measurement of reaction equilibrium and values of gef for the reactants and products
- Since

$$\frac{\Delta G^\circ_T}{T} = \frac{\Delta\left(G^\circ_T - H^\circ_{298}\right)}{T} + \frac{\Delta H^\circ_{298}}{T} = -R\ln K$$

$$\frac{\Delta H^\circ_{298}}{T} = -R\ln K - \frac{\Delta\left(G^\circ_T - H^\circ_{298}\right)}{T}$$

- Or

$$\Delta H^\circ_{298} = -RT\ln K - T\Delta\text{gef}$$

# Which method when?

| | Second law | Sigma method | Third law |
|---|---|---|---|
| **You need** | K at several T | K at several T + $c_p$ data | K (even one point) + gef tables |
| **You get** | Mean $\Delta H°$, $\Delta S°$ over the range | Full $\Delta G°(T)$ expression | $\Delta H°_{298}$ from every point |
| **Use it when** | Data are sparse | Good $c_p$ data exist | gef tables exist; best check for errors |

*Third law is generally considered the most reliable: it exposes systematic errors.*

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K$$

The goal of every method today: find $\Delta G^\circ$ (and $\Delta H^\circ$, $\Delta S^\circ$) of a reaction

## A Look it up

**Gibbs energy function (gef)**

Precise $\Delta G^\circ$ at any T from tabulated data

## B Measure K vs T

**Second- and third-law methods**

$\Delta H^\circ$ and $\Delta S^\circ$ from gas equilibria

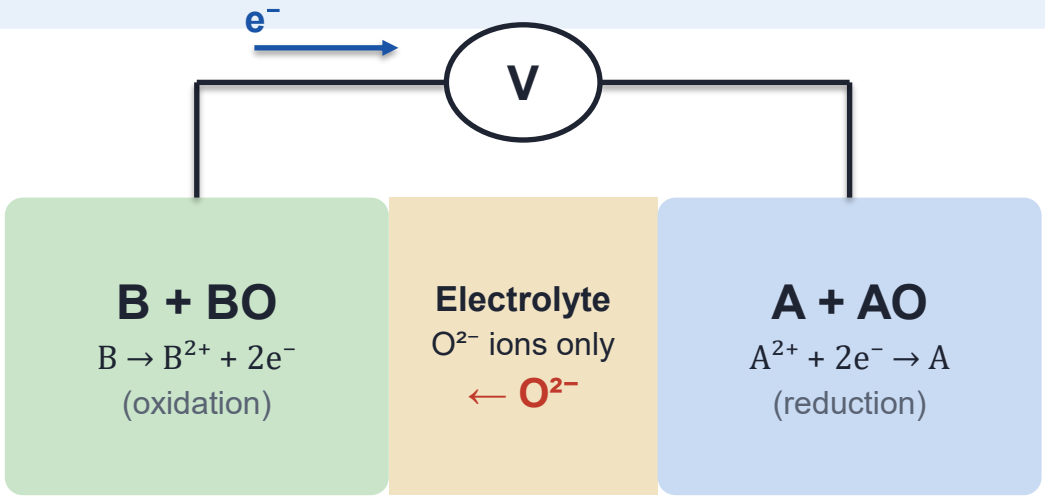
## C Measure a voltage

**Electrochemical (EMF) cells**

$\Delta G^\circ$ and activities read as a voltage E

# C · The idea: electrons through a wire

$$AO + B = BO + A$$




*"Electrochemical cell"*

**Electrons**
can only travel through the wire

**Ions**
travel through the electrolyte

**Voltage E**
at zero current measures the driving force: ΔG

# C · From voltage to Gibbs energy

$$\Delta G = -zFE$$

standard state: $\Delta G^\circ = -zFE^\circ$

- **z** — electrons transferred per reaction
- **F** — Faraday constant = 96 485 C/mol
- **E** — cell voltage at zero current (V)

**Quick example**

$z = 2,\ E = 0.50\ \text{V}$

$\Delta G = -2 \times 96\,485 \times 0.50$

$\mathbf{= -96.5\ kJ}$

**E > 0** — ΔG < 0: spontaneous

**E < 0** — ΔG > 0: not spontaneous

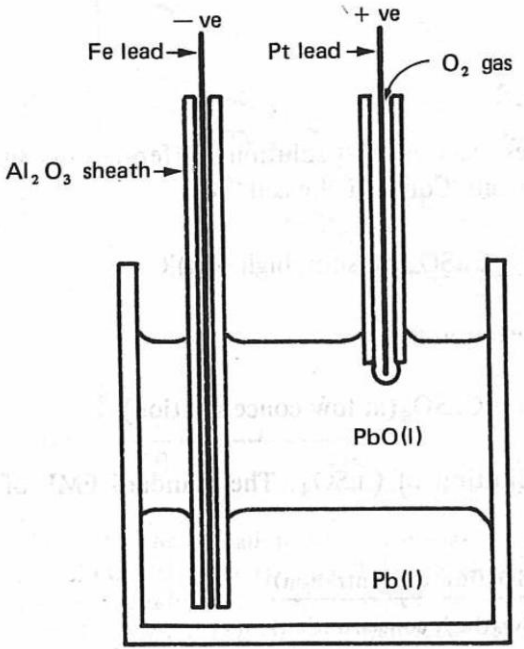
# C · Worked example: Pb–PbO formation cell

$$Pb(l) + \tfrac{1}{2}O_2(g) = PbO(l)$$

$$Pb(l) \mid PbO(l) \mid O_2(1\ atm), Pt$$

1. **Electrons per reaction**
   $Pb \rightarrow Pb^{2+} + 2e^-$ so $z = 2$
2. **ΔG° at 1100 K**
   $\Delta G^\circ = -191\,750 + 79.08 \times 1100$
   **$= -104.8$ kJ**
3. **Expected voltage**
   $E^\circ = -\Delta G^\circ / zF = 104\,762 / (2 \times 96\,485)$

**$E^\circ \approx 0.54$ V**




Sridhar & Jeffes, Trans. Inst. Min. Met. 1967
$\Delta G^\circ = -191\,750 + 79.08T$ J

# Can we measure Gibbs energy using a voltmeter?

# Electrochemical measurements

- A hypothetical reaction:

$$AO + B = BO + A$$

  - A changes oxidation state from +2 till 0
  - B changes oxidation state from 0 till +2
  - In other words,

$$A^{2+}O^{2-} + B^{0} = B^{2+}O^{2-} + A^{0}$$

# Electrochemical measurements

- Can write half-reactions that involve electrons

$$A^{2+} + 2e^{-} = A^{0}$$

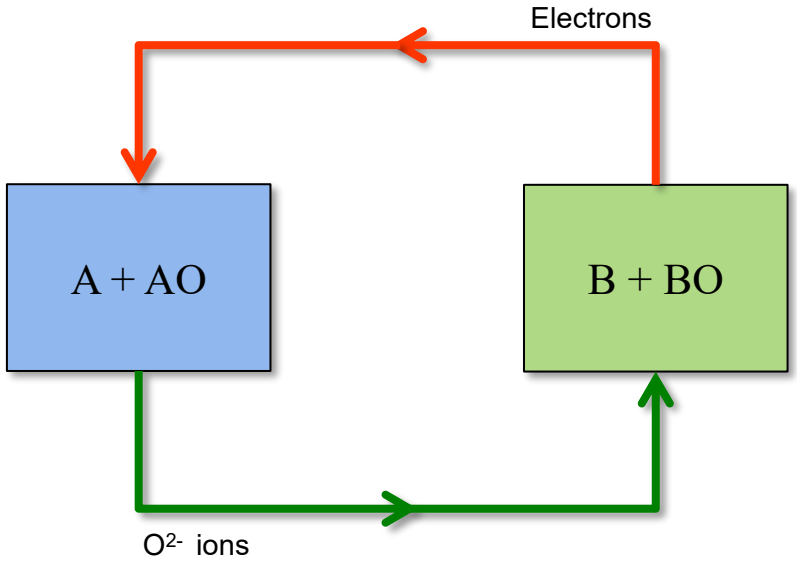
$$B^{0} = B^{2+} + 2e^{-}$$

- If the reaction is spontaneous
  - Gibbs energy minimizes, so $\Delta G$ must be related somehow to the electron transfer!

# Electrochemical measurements

- One can specially construct a system:

$$AO + B = BO + A$$




"Electrochemical cell"

- Two compartments, one containing A + AO, and the other with B + BO
  - Coupled together via 2 channels
- If this reaction is spontaneous, what happens?

1. B oxidizes and liberates electrons

$$B^0 = B^{2+} + 2e^-$$

2. A consumes electrons and is reduced, producing $O^{2-}$ ions

$$A^{2+} + 2e^- = A^0$$

# Electrochemical measurements

- Electromotive force (EMF)

$$\text{EMF} = \text{electromotive force [V]} = \frac{\text{work [J]}}{\text{charge [C]}}$$

  - 1 J work is produced when 1 C is transferred between two points with 1 V potential in electric circuit

$$E = \frac{w}{q} \longrightarrow w = qE$$

- Charge, $q$, is equal to ($z$ total moles of charge)

$$q = zF \longrightarrow w = zFE$$

  - $F$ ≡ Faradays constant = 96 485 C/mol electrons

# Electrochemical measurements

- Faradays constant
  - Charge of one mole protons, electrons

$$F = L \cdot e \qquad [\mathrm{C \cdot mol^{-1}}]$$

- $L$ ≡ Avogadros number ($6.023 \times 10^{23}$ particles/mol)
- $e$ ≡ elementary charge [Coulomb]

Coulomb is the SI unit
for electric charge
$1\mathrm{C} = 1\mathrm{A} \cdot 1\mathrm{s}$

$$F = 6.0221415 \times 10^{23} \cdot 1.6021765 \times 10^{-19}\,\mathrm{C}$$

$$F = 96\,485\ \mathrm{C \cdot mol^{-1}}$$



# Electrochemical measurements

- At constant *T* and *P*

$$-dG = \delta w'_{\max}$$ *w'* is non *P*–*V* work

  - Electrical work in this case

- At maximum cell potential, the difference in electrical work is

$$\delta w'_{\max} = FE_{\max} dz$$

$$dG = -FE_{\max} dz$$

# Electrochemical measurements

- Therefore

$$\Delta G = -zFE$$

- At the standard state,

$$\Delta G^\circ = -zFE^\circ$$

  - A positive cell potential, *E*, gives a negative $\Delta G$ (spontaneous)

# Electrochemical measurements

- For non-standard state:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

  - Therefore

$$-zFE = -zFE^\circ + RT \ln Q$$

$$E = E^\circ - \frac{RT}{zF} \ln Q$$ Nernst equation

  - For the reaction aA + bB = cC + dD

$$E = E^\circ - \frac{RT}{zF} \ln \left( \frac{a_C^c a_D^d}{a_A^a a_B^b} \right)$$

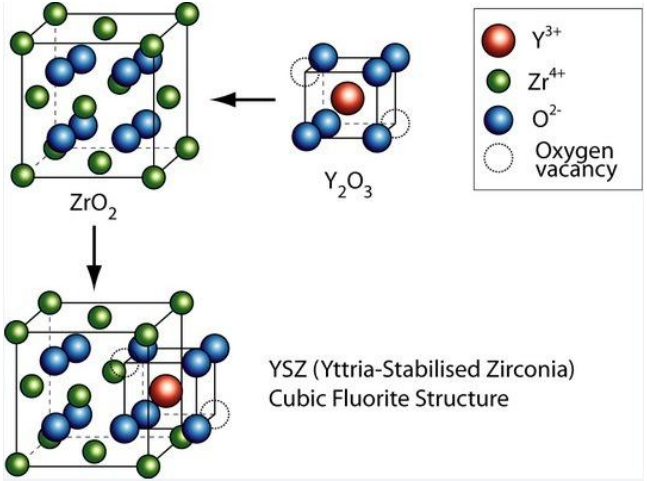


# Electrochemical measurements

- In many systems it is possible to construct galvanic cells to measure the emf, and thus derive thermodynamic properties
  - Needs a suitable electrolyte: only ionically conductive, cations with single valency state
  - Single electrode processes that occur reversibly at each electrode
  - At high temperatures, things can become very difficult
    - *Gas permeability, increased electronic conduction, materials compatibility, side reactions, etc.*

# Types of electrolytes, electrodes

- Liquid electrolytes
  - Aqueous
  - Molten oxides
  - Molten salts
- Solid electrolytes
  - Glass
  - Solid oxides
- Electrodes
  - Solid
  - Liquid
  - Gas




Zirconia is an ionic conductor

# Formation cells

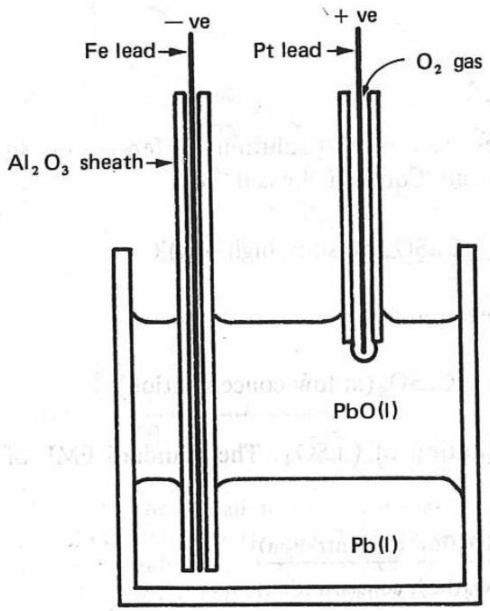
- Reaction:

$$Pb(l) + \frac{1}{2}O_2(g) = PbO\ (l)$$

$$Pb(l) \mid PbO(l) \mid O_2(Pt)$$

  - Anode: $Pb(l) = Pb^{2+}(oxide) + 2e^-$
  - Cathode: $\frac{1}{2}O_2(g) + 2e^- = O^{2-}(oxide)$
  - At constant 1 atm $O_2$, $\Delta G° = -2FE°$
    - > *Therefore $\Delta G°$ can be measured as a function of T*

$$\Delta G° = -191750 + 79.08 \cdot T \ [J]$$




Pb-PbO formation cell

Sridhar, Jeffes, *Trans. Inst. Min. Met.* 1967

# Formation cells

- Formation cell

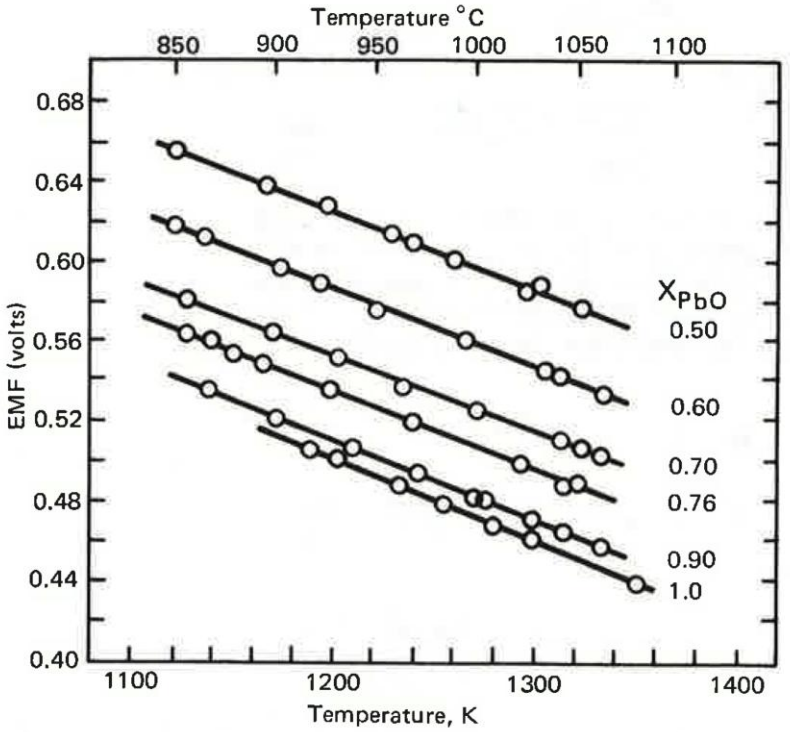
$$\mathrm{Pb(l) \mid PbO(l) \mid O_2(Pt)}$$

  - With the addition of a more stable oxide (ex. $SiO_2$ to the oxide phase

$$E = E^{\circ} - \frac{RT}{2F} \ln \left( \frac{a_{\mathrm{PbO(oxide)}}}{a_{\mathrm{Pb}} \cdot p_{O_2}^{1/2}} \right)$$

    > *Can measure the EMF as a function of PbO in the oxide phase*

$$\longrightarrow \quad a_{\mathrm{PbO}}$$




EMF as a function of temperature

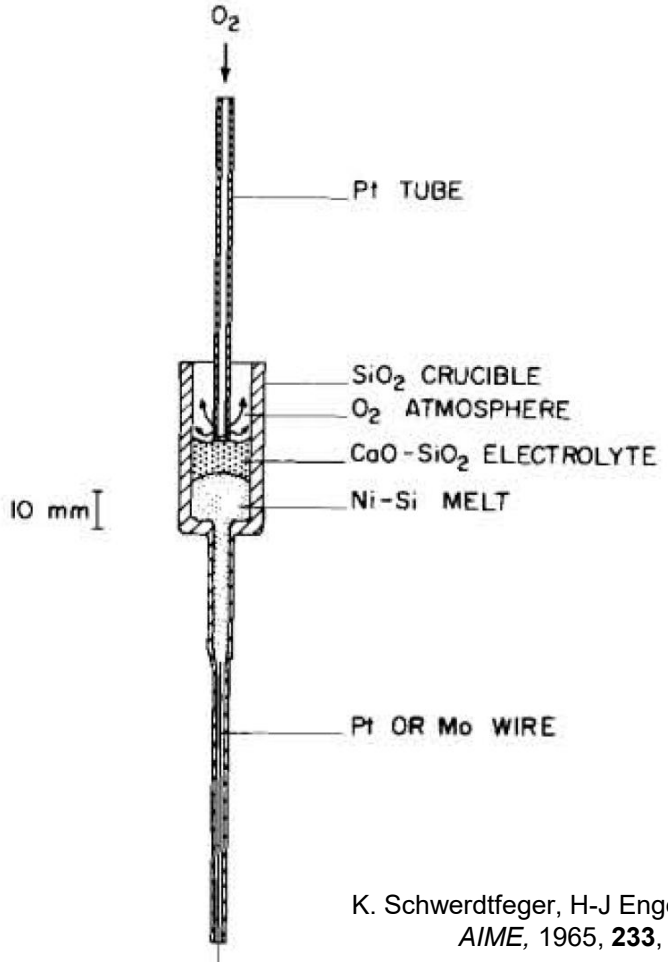
# Example

- Si–Ni system at 1480 °C (1753 K) to 1610 °C (1883 K)

$$Si(met) + O_2(g) \rightarrow SiO_2 \text{ (electrolyte)}$$

$$Si(l) \mid CaO–SiO_2(l), SiO_2(sat) \mid O_2(g)$$

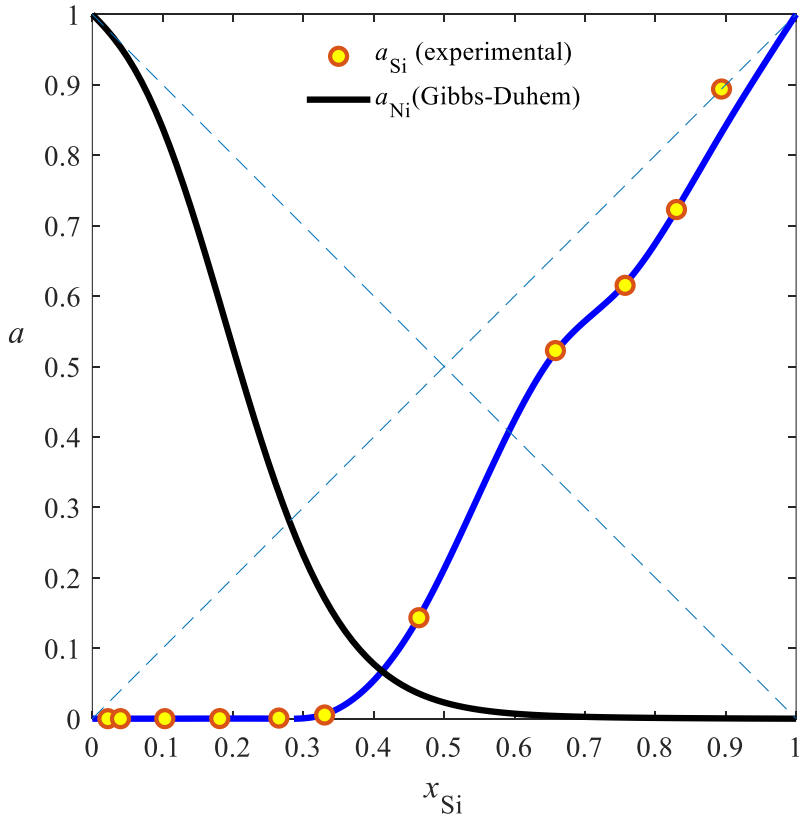
$$Ni–Si(l) \mid CaO–SiO_2(l), SiO_2(sat) \mid O_2(g)$$




K. Schwerdtfeger, H-J Engel , *Trans. Met. AIME,* 1965, **233**, 1327.

# Example

- Si–Ni system at 1480 °C (1753 K) to 1610 °C (1883 K)




Activities in the Si–Ni(l) system at 1883 K

# Concentration cells

- Identical electrodes, but in solutions with different concentrations

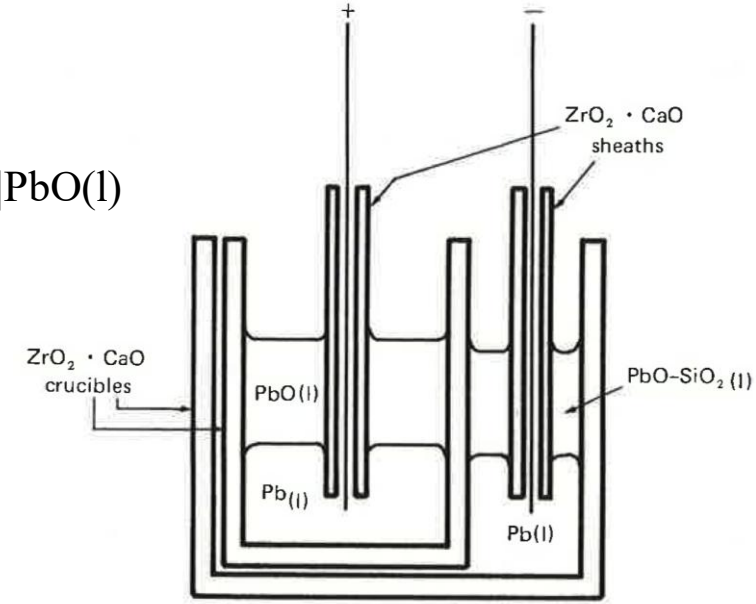
$$Pb(l) \mid PbO(\text{in } PbO - SiO_2, l) \mid CaO - ZrO_2 \mid Pb(l) \mid PbO(l)$$

  - Spontaneous process

$$PbO(l) \rightarrow PbO(\text{in } PbO - SiO_2, l)$$

  - Solid electrolyte ($ZrO_2$), passes $O^{2-}$ ions (ionically conductive)
  - For this cell,

$$E = -\frac{RT}{2F} \ln\left(\frac{a_{PbO(low)}}{a_{PbO(high)}}\right)$$




Pb-PbO concentration cell

Kozuka, Samis, *Met. Trans. B*, 1970

# Summary: three routes to ΔG°

## A Gibbs energy function

**Measure**

Look up gef and $\Delta H^\circ_{298}$

$$\Delta G^\circ_T = \Delta H^\circ_{298} - T\Delta gef$$

**You get**

Precise ΔG° at any T

## B Second & third law

**Measure**

Measure K vs T

$$\ln K = -\Delta H^\circ/RT + \Delta S^\circ/R$$

**You get**

ΔH°, ΔS° (third law: error check)

## C EMF cells

**Measure**

Measure voltage E

$$\Delta G = -zFE$$

**You get**

ΔG°, activities, oxygen probe

# References


1. R. A. Rapp, ed., Physicochemical Measurements in Metals Research, Part 1, 1970, Interscience Publishers.

2. D.R. Gaskell: Introduction to Metallurgical Thermodynamics, Second Edition, Hemisphere Publishing, 1981.